Business Intelligence & Analytics

The Comprehensive 12-Week Master Guide

Compiled for NPTEL Certification

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Week 1: Foundations of BI & Analytics

1. Understanding Intelligence & Decisions

In a business context, **Intelligence** is not just an abstract trait; it is a **tangible resource** or capability used to drive organizational success.

- **Informed Decisions:** Based on factual data, historical patterns, and statistical analytics. These minimize risk.
- **Uninformed Decisions:** Based on "gut feeling" or intuition. These are high-risk and difficult to replicate.

The Role of Hypothesis

Before analyzing data, we start with a **Conjecture/Hypothesis**—an argument or educated guess. We then collect and mine data specifically to prove or disprove this argument.

2. Modern Business Models

The Platform Model (Google): Unlike traditional businesses, Google acts as a platform connecting users to info/advertisers.

- **Curators:** These are the systems (or people) that ensure data quality. Without curation, a platform provides "noise" instead of "value."

3. The Long Tail Phenomena

Traditional retail follows the **80-20 Rule** (Pareto Principle), where 80% of revenue comes from the top 20% of popular products (the "Head"). Online businesses have broken this.

Example: The Power of Niche

The **"Head"** consists of a few mainstream hits (e.g., Marvel movies). The **"Long Tail"** consists of millions of niche products (e.g., indie documentaries). Individually, they sell little; combined, they can equal or exceed the "Head" revenue.

4. Data Infrastructure: Storage & Systems

To analyze data, we must first store and manage it correctly.

1.4.1 OLTP vs. Relational Technology

OLTP (Online Transaction Processing) is designed for day-to-day operations (like swiping a credit card). It led to the rise of **Relational Databases**, which allow for efficient storage and retrieval of massive datasets.

1.4.2 The Storage Hierarchy

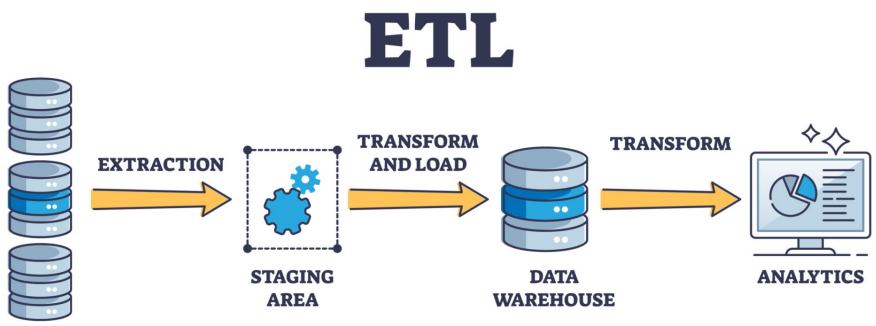
Technology	Purpose in BI
Database	Organized collection of structured data (the foundation).
Data Warehouse	A massive repository for *historical* data across the whole company.
Data Mart	A specific "slice" of the warehouse (e.g., just the Sales Department's data).
DSS	**Decision Support Systems**—tools that use this data to help managers decide.

5. The ETL Process: Preparing for Mining

Data in its raw form is usually "dirty." **ETL (Extraction, Transformation, Loading)** is the process that prepares it for analysis.

Process: The 3 Stages of ETL

1. **Extraction:** Pulling data from various sources (Big Data, online activity, sensors).
2. **Transformation:**
   - **Data Cleaning:** Removing "noise" and inconsistent data.
   - **Data Integration:** Merging data from different sources into one view.
   - **Data Transformation:** Aggregating data (e.g., daily sales → monthly totals).
3. **Loading:** Moving the clean, transformed data into the Data Warehouse.



6. Data Mining & Advanced Concepts

Data Mining is the extraction of hidden patterns using intelligent methods.

Exam Note: The rise of **ERPs** (Enterprise Resource Planning) is the major driver of data mining because it consolidated all business data into one place.

Big Data: Massive volumes of data generated online (high volume, velocity, and variety).

Descriptive Analysis: Summarizing past data to find trends (e.g., "Last year's sales grew by 10%").

Outliers: Data points that deviate from the norm.

Example: Spotting Outliers

In banking, if a user who usually spends $20/day suddenly spends $5,000 in a different country, that transaction is an **outlier**—likely fraud.